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WRECKER

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Chapter 1

OVERVIEW

1.1. Overview. This Qualification Training Package (QTP) establishes standardized training requirements and content for all Department of the Air Force (DAF) and Space Force personnel responsible for the safe operation, inspection, and preventative maintenance of Wrecker vehicles. Designed to ensure consistency and excellence across all installations, this package provides comprehensive guidance for both trainers and trainees throughout the qualification process.

1.1.1. The training standards outlined herein align with DAF requirements that supplement federal regulations. This package serves as the authoritative source for curriculum development and performance evaluation. Trainers will utilize Chapters 2 through 6 to guide trainee preparation, conduct classroom lectures, provide demonstrations, and administer knowledge and performance evaluations.

1.1.2. Trainees are required to read this entire lesson plan and the manufacturer's operator manual prior to starting lectures and must follow along with these materials during instruction. Successful completion of this training program demonstrates operator proficiency and supports mission readiness by maintaining the highest standards of safety, compliance, and operational effectiveness across both services.

1.2. Applicability. This QTP applies to all DAF personnel operating wreckers.

1.2.1. Trainees will complete a hands-on Performance Test, **Attachment 4**.

1.2.2. Upon successful completion, the trainer will document the results and sign off the respective tasks in the member's Individual Training Record (ITR), DAF Form 623, *Individual Training Record Folder*, or electronic On-the-Job Training (OJT) record.

1.3. Administrative. Send comments and suggested improvements on DAF Form 847, *Recommendation for Change of Publication*, through Air Force Installation and Mission Support Center (AFIMSC) functional managers via e-mail at: AFIMSC.A34OG.GroundTrans@us.af.mil.

Chapter 2

RESPONSIBILITIES

2.1. Trainer. Provide an overview of the training outlined in **Chapters 4** and **5**. Deliver classroom lectures and conduct trainee preparation using the lesson plans. Provide knowledge overview training, conduct demonstrations, and evaluate performance activities. Coordinate with the Unit Training Manager (UTM) or designated authority to ensure training is properly documented.

2.1.1. Conduct knowledge training with the trainees using this QTP, QTP 24-3, *General Vehicle Safety*, and the manufacturer's operator manual.

2.1.2. Conduct performance task explanations and demonstrations IAW **Chapter 6**.

2.1.3. Verify completion of Sections I and II on the Trainee's DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License*.

2.1.4. Document and sign-off the completed task(s) in OJT records.

2.2. Trainee. Read QTP 24-3 and this QTP in its entirety prior to the lecture. Complete the required performance test.

2.2.1. Follow along with the lectures using this lesson plan and its attachments.

2.2.2. Ensure the trainer explains the QTP process and individual responsibilities. Ask questions if they do not understand the objectives for each unit.

2.2.3. Must maintain a DAF Form 171 with Sections I and II completed, a valid state-issued driver's license, and a valid Common Access Card (CAC) on their person during training.

2.2.4. Successfully complete the hands-on performance test with zero instructor assists.

2.2.5. Use **Attachment 2** as a guide for vehicle inspections.

2.3. Certifier. The certifier acts as the independent evaluating authority. They are responsible for evaluating the Airman's physical task performance (**Chapter 7**) without providing any assistance or coaching. Upon successful, independent completion of the performance test, the certifier will sign off on the task(s) to finalize the licensing process.

Chapter 3

INTRODUCTION

3.1. Objectives. Upon completion of lectures, demonstrations, and hands-on sessions, trainees will successfully meet all training requirements. This training ensures the trainee becomes a qualified operator who possesses the knowledge and skills to properly operate the vehicles applicable to this package. Trainees must complete all components of the performance test with zero instructor assists.

3.1.1. Train and qualify each trainee in safe operation and preventive maintenance of the wrecker.

3.1.2. This training will ensure the trainee becomes a qualified operator; an operator who has the knowledge and skills to operate a wrecker in a safe, proficient and professional manner.

3.2. Desired Learning Outcome.

3.2.1. Understand the purpose of the wrecker and its role in the mission.

3.2.2. Understand the safety precautions to be followed for pre-, during- and post-operation inspections of the wrecker.

3.2.3. Know the proper operator maintenance procedures of the wrecker IAW applicable manual(s) and use of DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.2.4. Be completely familiar with the safety features of the wrecker.

3.2.5. Safely and proficiently operate the wrecker.

3.3. Lesson Duration. Recommended instructional and hands-on training time is 60 hours. **Note:** This is a recommended baseline; training time may vary depending on how quickly a trainee demonstrates proficiency in the required tasks.

Table 3.1. Recommended Training Time for Training Activities.

Training Activity	Training Time
Trainee's Preparation	2 Hours
Instructor's Lecture	2 Hours
Instructor's Demonstration	24 Hours
Trainee's Personal Experience (to build confidence and proficiency) ▪ Perform Operator Maintenance ▪ Operate the Vehicle	30 Hours
Trainee's Performance Evaluation	2 Hours

3.4. Instructional References.

3.4.1. DAF Form 1800.

3.4.2. Risk Management (RM) and Safety Principles.

3.4.3. Applicable TOs or Manufacturer's Operator's Manual (see Vehicle Management for TO number for vehicle being used in training).

3.4.4. United States Department of Transportation, Federal Motor Carrier Safety Administration; on-line at http://www.fmcsa.dot.gov/.

3.4.5. Special references based-off type of vehicle.

3.5. Instructional Training Aids and Equipment.

3.5.1. Wrecker and General Vehicle Safety QTPs.

3.5.2. Wrecker.

3.5.3. Applicable TO or Manufacturer's Operator's Manual.

3.5.4. DAF Form 1800.

3.5.5. Videos (if locally produced).

3.5.6. Suitable training area.

3.5.7. Various types of vehicles to be towed.

3.5.8. Traffic cones.

Chapter 4

KNOWLEDGE

4.1. Knowledge Overview. Recovery vehicles are commonly referred to as tow trucks, wreckers, or rollbacks. For the purpose of this QTP, the term "wrecker" will be utilized. To support diverse DAF missions, various types of wreckers are employed to accomplish vehicle recovery operations.

4.1.1. Scope and Applicability. Wrecker operators require specialized training and licensing to ensure safe and proficient operation. Operators must be prepared to execute vehicle recoveries, manage accident scenes involving overturned vehicles, and handle cargo shifts or spills on roadways.

4.1.2. Each wrecker type possesses unique operating characteristics for safely retrieving disabled vehicles. This chapter provides foundational knowledge for wrecker operations; however, operators must thoroughly review and comply with the manufacturer's operator's manual for the specific wrecker being utilized.

4.2. Wrecker Design. A wrecker is a truck equipped with a permanently affixed hoist, towing apparatus, self-loading flatbed, or any combination of these items used to transport disabled vehicles. Basic Wrecker Types:

4.2.1. Rollback: Are ideal for most modern vehicles. They prevent damage to the underside of the vehicle and are the standard choice for accident vehicles that have been totaled.

Figure 4.1. Rollback.



4.2.2. Hook: Required for special purposes and heavy vehicles (e.g., tractor-trailers).

Figure 4.2. Hook.



4.2.3. Wheel-lift: Light and Medium-duty wreckers utilize wheel-lifts for passenger cars and small trucks.

Figure 4.3.Wheel-lift.



4.2.4. Dual-Purpose: Some wrecker designs allow the vehicle to retrieve assets using either the wheel-lift or hook method.

4.2.5. Flatbed Towing: Certain flatbed wreckers are designed to tow an additional vehicle using an attached wheel-lift or hook attachment on the rear.

4.3. Wrecker Specifications and Components.

4.3.1. Capacity and Data Plates. Information concerning the safe load capacity of a wrecker is located on its manufacturer data plate or in the appropriate Technical Order (TO). The data plate specifications dictate the boom ratings, winch ratings, wire rope specifications, and under-lift ratings (rated loads/capabilities).

4.3.2. Common Truck Components. The towing portion of a wrecker is often manufactured by a different company than the truck chassis. Common integrated components include:

4.3.2.1. Air Brakes. Standard heavy-duty braking systems, covered in **Chapter 5.**

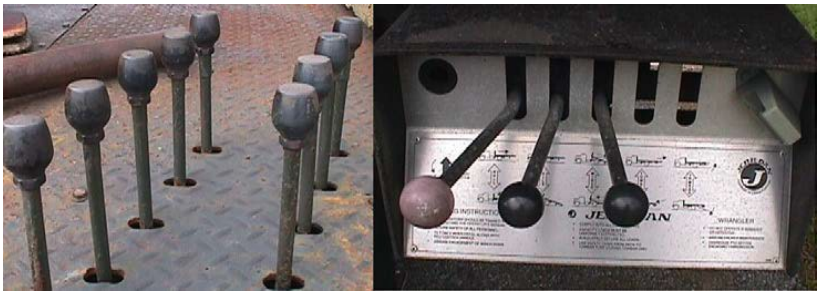
4.3.2.2. Power Take-Off (PTO). The PTO control is located in the cab of the truck (dashboard or floor). When engaged, it supplies engine power to a hydraulic pump that operates the hoist cables, tow bar assembly, or flatbed extensions.

4.3.2.2.1. Ensure the PTO is completely disengaged while the vehicle is traveling.

4.3.3. Towing Assembly Components.

4.3.3.1. Hoist Control Levers. The control point for raising, lowering, and extending the tow bar, wheel-lift, flatbed, and cable hoists.

Figure 4.4. Hoist Control Levers.



4.3.3.2. Throttle Control Lever. Controls the engine idle speed from the rear of the wrecker. Heavier vehicles require a faster wrecker idle speed to operate the hydraulics effectively.

4.3.3.2.1. CAUTION: Never drive or operate the wrecker in transit with the throttle control lever engaged.

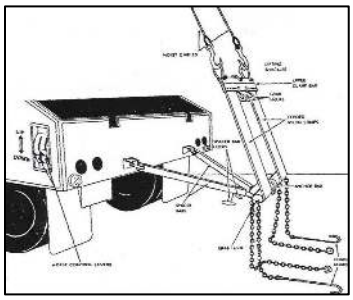
4.3.3.3. Hoist Cables and Cable Spools. Cables operate the lifting components, pull vehicles onto flatbeds, and execute snatch block operations.

4.3.4. Warning Lights (Beacons). Wreckers feature upper-mounted beacon lights. During recovery operations, these lights warn oncoming motorists and serve as a critical safety device for the operator.

4.3.5. Tow Bar Assembly and Booms (Hook Wreckers).

4.3.5.1. Sling Tow Bar. Uses a spacer to maintain a positive distance between the towed vehicle and the wrecker. Chain hooks secure the rear axle of the disabled vehicle to the wrecker's tow bar before lifting.

Figure 4.5. Sling Tow Bar Assembly.



4.3.5.2. Booms. The boom supports the weight of the towed vehicle. Booms are adjustable; heavier loads require higher boom angles (usually 30-45 degrees). Raising the boom places more weight on the wrecker's front wheels.

4.3.5.2.1. CAUTION: Never raise or lower the boom when a vehicle is attached to the tow bar assembly. Doing so can cause severe injury and damage the equipment.

4.3.6. Outriggers. Stabilizing arms located just behind the cab on certain wreckers. They anchor the wrecker during heavy or side-angle recovery operations.

4.4. Physics Application and Recovery Equipment.

4.4.1. Scotch Blocks (Rear Wheel Chocks). Used in conjunction with parking brakes to anchor the wrecker during snatch block recovery operations.

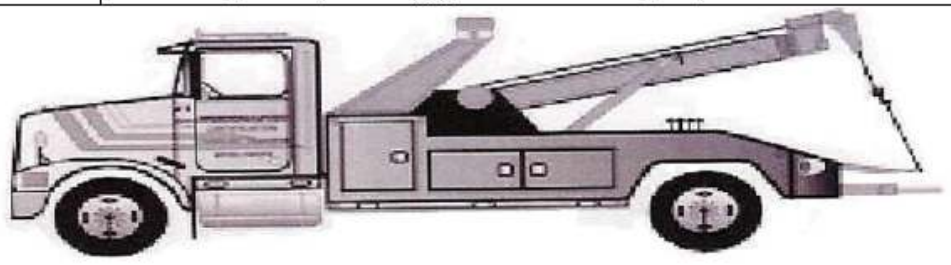
Figure 4.6. Scotch Blocks.



4.4.2. Towing Capacity Formulas. Operators must never exceed Gross Axle Weight Rating (GAWR), Gross Combination Weight Rating (GCWR), or Gross Vehicle Weight Rating (GVWR). To calculate capability, refer to the manufacturer data plate and utilize the standard overhang formulas.

Figure 4.7. Towing Capacity Formulas.

Variable	Description
A	Weight of the truck, driver, fuel, and all equipment on any conventional weight scale divided by 2.
B	Measure the distance between the centers of the front axle and the rear axle.
C	Measure the distance from the center of the rear axle to the towing position of the tow bar or cross arm.
D	Multiply half wrecker weight (A) by wheelbase (B). The sum is known as inch pounds. Divide inch pounds by overhang (C). Sum is the tow capacity of truck.



A [ ] X B [ ] = C [ ] / [ ]

1/2 Front Axle Weight | Wheel Base (inches) | Divided by Overhang (inches)

D [ ] Towing Capacity (T.C.)

4.4.3. Scene Survey and Emergency Warning Devices. Operators must secure the work area prior to recovery. Place warning triangles or flares utilizing extreme caution:

4.4.3.1. Two-lane road: 10 ft front, 10 ft rear, and 100 ft rear.

4.4.3.2. Curves/Hills: 10 ft rear and 300 ft rear.

4.4.3.3. Divided highway: 10 ft, 100 ft, and 200 ft to the rear.

4.4.4. Hazardous Materials. Immediately clear the scene if hazardous materials (e.g., explosive fluids, chemical spills) are present. Proceed only when authorized by the Incident Commander or appropriate authorities.

4.4.5. Hoist Cables and Cable Spools.

4.4.5.1. Crossover. Uneven winding causes excessive cable abuse. Do not allow the cable to cross over itself; keep it uniform on the drum. *Tip: Place a weight on the hook when rewinding to maintain tension.*

4.4.5.2. Freewheeling. When letting out large amounts of cable, release the freewheeling device and pull the cable manually to prevent drum crossover.

4.4.6. Chains, Hooks, and Snatch Blocks.

4.4.6.1. Chains. Only utilize Grade 7 Transport Chains (for towing) or Grade 8 Alloy Chains (for recovery).

4.4.6.1.1. WARNING: Working Load Limit (WLL) is the maximum uniform tension applied to a straight, new chain. Exceeding limits poses a fatal risk. Operators must inspect chains before every use.

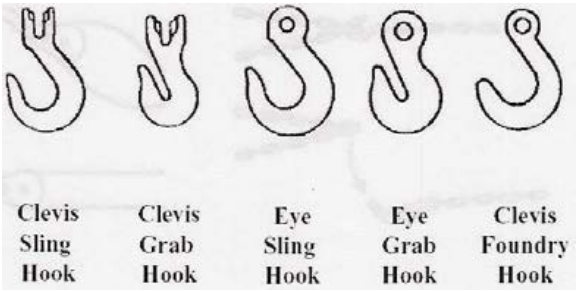
Table 4.1. Recommended Towing/Recovery Chain types.

Grade 7 Transport Chain (Towing)			**Grade 8 Alloy Chain (Recovery)**		
Chain Size (in.)	**Weight per 10 ft. (lbs.)**	**Working Load Limit (lbs.)**	**Chain Size (in.)**	**Weight per 10 ft. (lbs.)**	**Working Load Limit (lbs.)**
1/4	8.1	3,150	9/32 (1/4)	7.2	3,500
5/16	9.8	4,700	5/16	11.4	5,400
3/8	14.1	6,600	3/8	14.8	7,100
7/16	21.6	8,750	1/2	24.3	12,000
1/2	24.6	11,300	5/8	35.1	18,100
			3/4	58.4	28,300
			7/8	70.5	34,200
5/8 Grade 8 – Heavy recovery ~ 1/2 Grade 8 – Medium recovery ~ 3/8 Grade 8 – Light recovery.					

Note: CAUTION – Working Load Limit: This is the maximum load in pounds, which should ever be applied to the chain, when the chain is in new or like new condition. This rating applies when the load is uniformly applied, in direct tension, to a straight length of chain. The operator puts themselves and others at risk of injury or death anytime they exceed limits. Ensure he/she inspect chains before each use.

4.4.6.2. Hooks. Eye grab hooks and clevis grab hooks are the standard attachments for recovery operations.

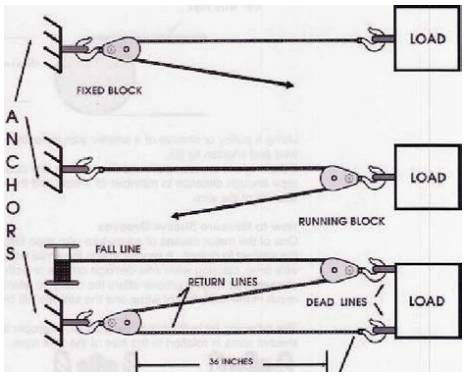
Figure 4.8. Hook Types.



4.4.6.3. Snatch Blocks. A pulley with a hook on one end used to increase winching capability. Do not winch two snatch blocks closer together than 36 inches.

4.4.6.3.1. WARNING: Misused or poorly maintained snatch blocks present severe fatal hazards. Always place hooks so the back of the hook faces the ground or away from personnel (if a hook fails, it snaps in the direction of its back).

Figure 4.9. Snatch Block Configurations.



4.4.7. General Tow Procedures.

4.4.7.1. Do not tow a vehicle on its drive wheels without protecting the transmission. Use a tow dolly or remove the drive shaft (mechanic recommended).

4.4.7.2. When towing from the rear, secure the steering wheel with the front wheels pointing straight using a specialized strap/hook device.

4.4.7.3. Always utilize two independent safety chains on every tow.

4.5. Vehicle Inspection.

4.5.1. Pre-Trip Inspection. Operators must inspect the wrecker using DAF Form 1800 and the Seven-Step Inspection Method. Report any discrepancies to the Vehicle Control Official (VCO) or maintenance immediately. Key inspection points include:

4.5.1.1. Approach: General condition, exterior damage, fluid leaks.

4.5.1.2. Wheels/Tires: Rim damage, proper air pressure, tight lug nuts, valve stems, proper tread depth (4/32 front, 2/32 rear).

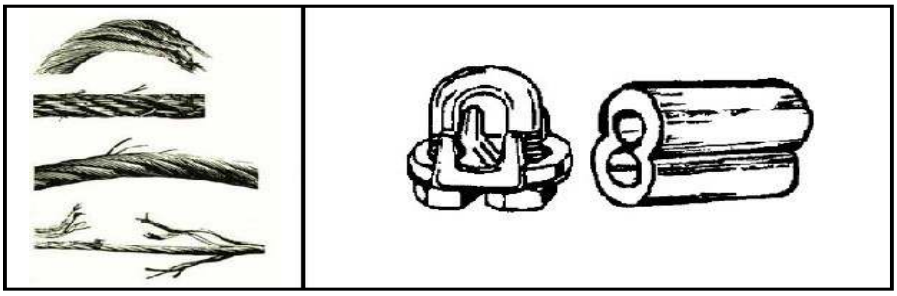
4.5.1.3. Engine Compartment: Fluid levels (oil, coolant, power steering, transmission), belt tension (max ¼ inch movement), battery terminal corrosion, and exposed wiring.

4.5.1.4. Suspension/Steering: Cracked spring hangers, missing leaves, leaking shocks, steering wheel play (max 2 inches).

4.5.1.5. Recovery Components: Booms, wheel-lifts, outriggers, pulleys, and winch cables.

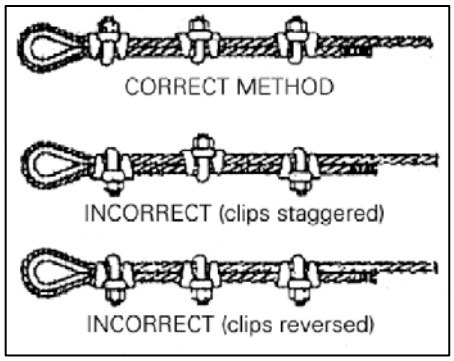
4.5.2. Winch Cable (Wire Rope) Inspection. Cables must be immediately removed from service if damaged. Inspect for proper lubrication, frays, cuts, crushing, kinks, or "bird caging" (twisting of the rope in the opposite direction of the lay).

Figure 4.10. Cable Damage and U-bolts/Wire Clips.



4.5.2.1. Cable Fasteners. Ensure U-bolt wire clips and compression sleeves are correctly installed. The saddle of the clip must rest against the "live" portion of the rope, while the U-bolt rests on the "dead" (short) end. *Rule of thumb: "Never saddle a dead horse."*

Figure 4.11. Methods for U-Bolt Wire Clips.



4.5.3. Emergency Equipment. Ensure the wrecker is equipped with a charged fire extinguisher, warning triangles/flares, a fully stocked first aid kit, and towed vehicle light attachments.

4.6. Vehicle Safety and Operator Equipment.

4.6.1. Hazard Awareness. Operators must remain vigilant of moving traffic, crush points between the wrecker and the disabled vehicle, and the severe danger zone around tensioned cables.

4.6.2. Protective Equipment. Operators must wear impact-resistant safety footwear (American Society for Testing and Materials (ASTM) F2412/F2413 compliant), heavy-duty work gloves (remove rings/jewelry), a reflective vest, and appropriate weather/eye/hearing protection.

4.6.3. Passenger Restrictions. Personnel are only permitted to ride in the cab of the wrecker, provided they wear a seatbelt. Never allow anyone to ride inside the towed vehicle.

4.7. Driving Safety and Precautions.

4.7.1. General Safety.

4.7.1.1. Plan routes to minimize backing. If backing is required, utilize a spotter QTP24-3, *General Vehicle Safety*. Stop immediately if you lose sight of the spotter.

4.7.1.2. Never exceed load limits or boom ratings. Keep the towed load within one foot of the ground whenever possible.

4.7.1.3. Never get under a disabled vehicle raised solely by the wrecker. Utilize adequate safety blocks or jack stands.

4.7.2. Tire Changing Safety.

4.7.2.1. Never change a tire on a bridge, sharp curve, or roadway lacking a shoulder. Ensure the surface is firm and level.

4.7.2.2. Turn on four-way flashers, place warning triangles, block the opposite wheels, and apply the parking brake.

4.7.2.3. Ensure the jack is properly rated and placed at the manufacturer's recommended lift point. Never get under the vehicle once lifted.

4.8. Wrecker Vehicle Operation.

4.8.1. Accelerating and Steering.

4.8.1.1. Accelerate gradually. Jerky movements can cause mechanical failure or drop the towed vehicle.

4.8.1.2. Utilize the hand-over-hand steering method. Account for the significantly longer turning radius created by the attached vehicle.

4.8.2. Shifting Gears and Braking.

4.8.2.1. Downshift before entering curves or descending steep grades to prevent brake fade and maintain control.

4.8.2.2. Maintain a wide space cushion. Due to the extreme combined weight, follow the three-second rule (or greater) to account for increased perception, reaction, and braking distances.

4.8.3. Flightline Operations. If operating on the flightline, strictly adhere to DAFI 13-213, *Airfield Driving*, and local airfield driving instructions.

Chapter 5

AIR BRAKES

5.1. Safe Operation of a CMV with an Air Brake System. Safe operation of a commercial motor vehicle (CMV) with an air brake system, requires knowledge of: Air Brake System nomenclature; the dangers of contaminated air supply; implications of severed or disconnected air lines between the power unit and the vehicle implications of low air pressure readings; procedures to conduct safe and accurate pre-trip inspections; procedures for conducting enroute and post-trip inspections of air actuated brake systems (ability to detect defects, which may cause the system to fail).

5.1.1. Air brakes are really three different braking systems:

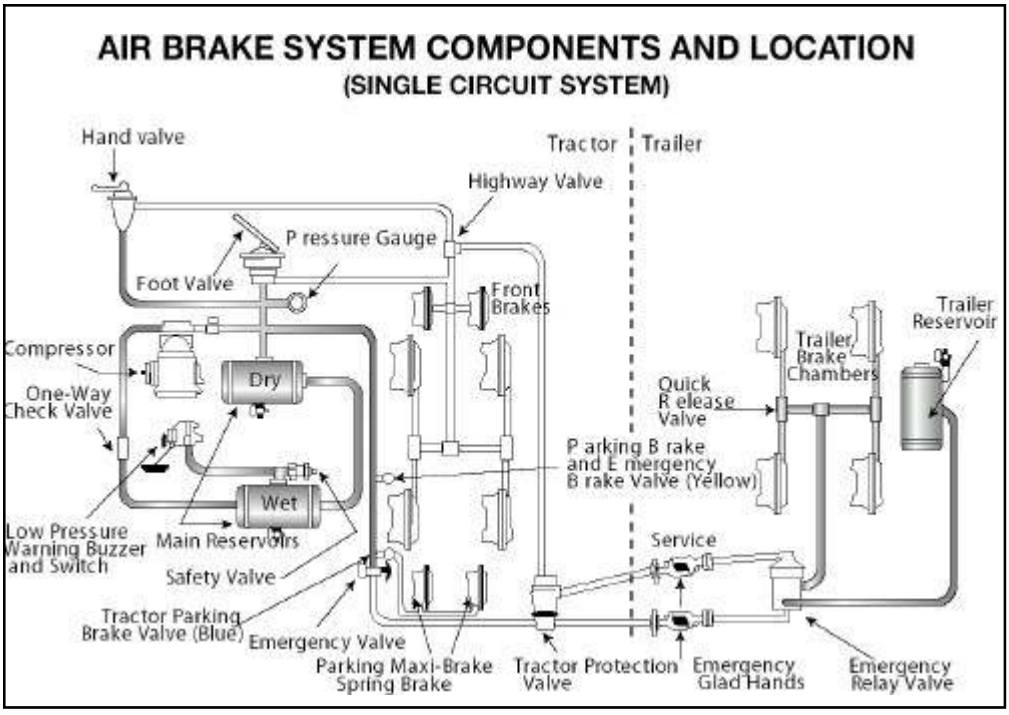
5.1.1.1. Service brake. Applies and releases the brakes when the operator uses the brake pedal during normal driving.

5.1.1.2. Parking brake. Applies and releases the parking brakes when the operator uses the parking brake control.

5.1.1.3. Emergency brake. Uses part of the service and parking brake systems to stop the vehicle in the event of a brake system failure.

5.1.2. Parts of an Air Brake System:

Figure 5.1. Air Brake System Components.



5.1.2.1. Air compressor. Pumps air into the air storage tanks (reservoirs). The air compressor is connected to the engine through gears or a v-belt. The compressor may be air cooled or may be cooled by the engine cooling system. It may have its own oil supply or be lubricated by engine oil. If the compressor has its own oil supply, check the oil level before driving.

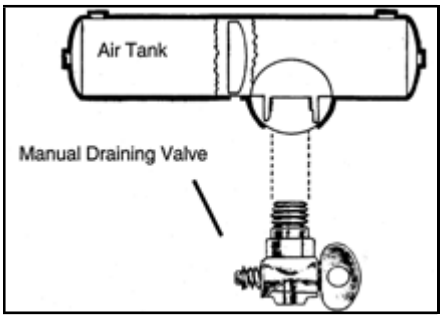
5.1.2.2. Air compressor governor. The governor controls when the air compressor will pump air into the air storage tanks. When air tank pressure rises to the "cut-out' level (around 125 pounds per square inch or psi), the governor stops the compressor from pumping air. When the tank pressure falls to the "cut-in" pressure (around 100 psi), the governor allows the compressor to start pumping again.

5.1.2.3. Air storage tanks. Air storage tanks are used to hold compressed air. The number and size of air tanks varies among vehicles. The tanks will hold enough air to allow the brakes to be used several times, even if the compressor stops working.

5.1.2.4. Air tank drains. Compressed air usually has some water and some compressor oil in it which is bad for the air brake system. The water can freeze in cold weather and cause brake failure. Drain the air tanks completely. Each air tank is equipped with a drain valve in the bottom. There are two types:

5.1.2.4.1. Manually operated. Turn a quarter turn or pull a cable to operate. See **Figure 5.2.** Manually drain the tank at the end of each day of driving.

Figure 5.2. Manual Drain Valve.



5.1.2.4.2. Automatic. The water and oil is automatically expelled. They may be equipped for manual draining as well. Automatic air tanks are available with electric heating devices to help prevent freezing of the automatic drain cold weather.

5.1.2.5. Alcohol evaporator. Some air brake systems have an alcohol evaporator to put alcohol into the air system. This helps to reduce the risk of ice in air brake valves and other parts during cold weather which can make the brakes stop working. Check and fill the alcohol container as necessary, every day during cold weather.

5.1.2.6. Safety valve. A safety relief valve is installed in the first tank the air compressor pumps air to. The safety valve protects the tank and the rest of the system from too much pressure. The valve is usually set to open at 150 psi. If the safety valve releases air,

something is wrong.

5.1.2.7. The brake pedal. Put on the brakes by pushing down the brake pedal. (It is also called the foot valve or treadle valve.)

5.1.2.7.1. Pushing the pedal down harder applies more air pressure. Letting up on the brake pedal reduces the air pressure and releases the brakes. Releasing the brakes releases some compressed air go out of the system, so the air pressure in the tanks is reduced.

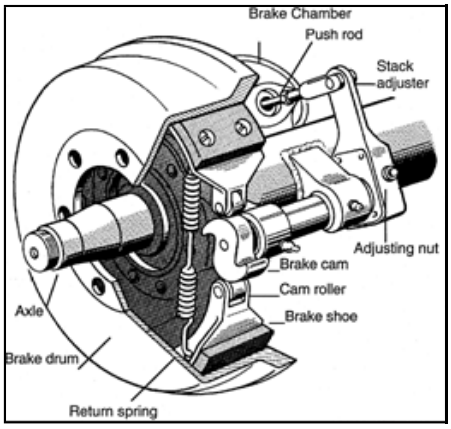
5.1.2.7.2. It must be made up by the air compressor. Pressing and releasing the pedal unnecessarily can release air out faster than the compressor can replace it. If the pressure gets too low, the brakes won't work.

5.1.3. Foundation brakes. Used at each wheel. The most common type is the s-cam drum brake, shown in **Figure 5.3.** Its parts include:

5.1.3.1. Brake drums, shoes, and linings. Located on each end of the vehicle's axles. The wheels are bolted to the drums. The braking mechanism is inside the drum. To stop, the brake shoes and linings are pushed against the inside of the drum. This causes friction, which slows the vehicle (and creates heat). The heat a drum can take without damage depends on how hard and how long the brakes are used. Too much heat can make the brakes stop working.

5.1.3.2. S-cam brakes. When the brake pedal is pushed, air is let into each brake chamber Air pressure pushes the rod out, moving the slack adjuster, thus twisting the brake camshaft. This turns the s-cam (so called because it is shaped like the letter "S"). The s-cam forces the brake shoes away from one another and presses them against the inside of the brake drum. When releasing the brake pedal, the s-cam rotates back and a spring pulls the brake shoes away from the drum, letting the wheels roll freely again.

Figure 5.3. S-cam Air Brakes.



5.1.3.3. Wedge brakes. The brake chamber push rod pushes a wedge directly between the ends of two brake shoes. This shoves them apart and against the inside of the brake drum. Wedge brakes may have a single brake chamber, or two brake chambers, pushing wedges

in at both ends of the brake shoes. Wedge type brakes may be self-adjusting or may require manual adjustment.

5.1.3.4. Disc brakes. In air-operated disc brakes, air pressure acts on a brake chamber and slack adjuster, like s-cam brakes. But instead of the s-cam, a "power screw" is used. The pressure of the brake chamber on the slack adjuster turns the power screw. The power screw clamps the disc or rotor between the brake lining pads of a caliper, similar to a large c-clamp.

5.1.4. Supply pressure gauges. All air-braked vehicles have a pressure gauge connected to the air tank. If the vehicle has a dual air brake system, there will be a gauge for each half of the system. (Or a single gauge with two needles.) Dual systems will be discussed later. These gauges tell the operator how much pressure is in the air tanks.

5.1.5. Application pressure gauge. This gauge shows how much air pressure is being applied to the brakes. (This gauge is not on all vehicles.) Increasing application pressure to hold the same speed means the brakes are fading. Slow down and use a lower gear. The need for increased pressure can also be caused by brakes out of adjustment, air leaks, or mechanical problems.

5.1.6. Low air pressure warning. A low air pressure warning signal is required on vehicles with air brakes. A warning signal that can be seen must come on before the air pressure in the tanks falls below 55 psi. (Or one half the compressor governor cutout pressure on older vehicles.) The warning is usually a red light. A buzzer may also come on.

5.1.6.1. Another type of warning is the "wig wag." This device drops a mechanical arm into the view when the pressure in the system drops below 55 psi. An automatic wig wag will rise out of the view when the pressure in the system goes above 55 psi. The manual reset type must be placed in the "out of view" position manually. It will not stay in place until the pressure in the system is above 55 psi.

5.1.7. Stop light switch. Drivers behind the vehicle must be warned when the brakes are put on. The air brake system does this with an electric switch that works by air pressure. The switch turns on the brake lights when the air brakes are put on.

5.1.8. Front brake limiting valve. Some older vehicles (made before 1975) have a front brake limiting valve and a control in the cab. The control is usually marked "normal" and "slippery." When the control is in the "slippery" position, the limiting valve cuts the "normal" air pressure to the front brakes by half.

5.1.8.1. Limiting valves were used to reduce the chance of the front wheels skidding on slippery surfaces. However, they actually reduce the stopping power of the vehicle.

5.1.8.2. Front wheel braking is good under all conditions. Tests have shown front wheel skids from braking are not likely even on ice. Make sure the control is in the "normal" position to have normal stopping power.

5.1.8.3. Many vehicles have automatic front wheel limiting valves. They reduce the air to

the front brakes except when the brakes are put on very hard (60 psi or more application pressure). These valves cannot be controlled by the driver.

5.1.9. Spring brakes. Tractor and straight truck spring brakes will come fully on when air pressure drops to a range of 20 to 45 psi (typically 20 to 30 psi). Do not wait for the brakes to come on automatically. When the low air pressure warning light and buzzer first come on, bring the vehicle to a safe stop right away, while the brakes can still be controlled. Brakes must be properly adjusted to work correctly and to prevent the braking power from being affected.

5.1.10. Parking brake controls. In newer vehicles with air brakes, engage the parking brakes using a diamond-shaped, yellow, push-pull control knob. Pull the knob out to put the parking brakes (spring brakes) on, and push it in to release them. On older vehicles, the parking brakes may be controlled by a lever. Use the parking brakes whenever the vehicle is parked. **Note:** Never push the brake pedal down when the spring brakes are on. The brakes could be damaged by the combined forces of the springs and the pressure.

5.1.10.1. Modulating control valves. A control handle on the dash board found in some vehicles that may be used to apply the spring brakes gradually. It is spring loaded so the operator has a feel for the braking action. The more the control lever is moved, the harder the spring brakes come on, so that the operator can control the spring brakes if the service brakes fail. When parking a vehicle with a modulating control valve, move the lever as far as it will go and hold it in place with the locking device.

5.1.10.2. Dual parking control valves. A dual valve is used when main air pressure is lost to release the spring brakes so the vehicle can be moved in an emergency. A push-pull type valve used to put the spring brakes for parking, and a spring loaded valve in the “out” position. When pushing the control in, air from the separate air tank releases the spring brakes so the vehicle can move. When releasing the button, the spring brakes come on again. There is only enough air in the separate tank to do this a few times. Plan carefully when moving or the vehicle may be stopped in a dangerous location when the separate air supply runs out.

5.1.10.3. Antilock Braking System. Truck tractors with air brakes built on or after March 1, 1997, and other air brakes vehicles, (trucks, buses, trailers, and converter dollies) built on or after March 1, 1998, are required to be equipped with antilock brakes. ABS is a computerized system that keeps the wheels from locking up during hard brake applications.

5.1.10.3.1. On newer vehicles, the malfunction lamp comes on at start-up for a bulb check, and then goes out quickly. On an older system, the lamp could stay on until the vehicle is driving over 5 miles per hour (MPH).

5.1.10.3.2. If the lamp stays on after the bulb check, or goes on once the vehicle is under way, the operator may have lost ABS control at one or more wheels.

5.1.10.3.3. ABS is in addition to the normal brakes. It does not decrease or increase

the normal braking capability. It only activates when wheels are about to lock up.

5.1.11. Dual air brakes. Two separate air brake systems which use a single set of brake controls. Each system has its own air tanks, hoses, lines, etc. Both systems supply air to the trailer (if there is one).

5.1.11.1. One system (the "Primary") typically operates regular brakes on rear axle or axles.

5.1.11.2. The other system (the "Secondary") operates regular brakes on front axle (and possibly one rear axle).

5.1.11.3. Before driving a vehicle, allow time for the air compressor to build up a minimum of 1000 psi pressure in both the primary and secondary systems, or to the recommended psi in the Manufacturer's Operator's Manual. **Note:** Pay attention to the low air pressure warning light and buzzer. Never operate if the warning light or buzzer is on (typically when the air pressure drops below 660 psi). Bring the vehicle to a stop right away and safely park the vehicle if this happens while driving.

5.1.12. Inspecting air brakes, there are added items to be included in Step 2 "Engine Compartment Checks", Step 5 "Walk-Around Inspecting", and Step 7 "Final Air Brake Check" of the Seven-Step Inspection Process. See **Attachment 5**. **Note:** If any of the tests fail, they must be reported to the VCO, the supervisor, and vehicle maintenance.

5.2. Using Air Brakes.

5.2.1. Normal stops. Push the brake pedal down. Control the pressure so the vehicle comes to a smooth, safe stop. If the vehicle has a manual transmission, do not push the clutch in until the engine revolutions per minute (RPM) is down close to idle. When stopped, select a starting gear.

5.2.2. Emergency stops. When stopping in an emergency situation, the operator should brake in a way that will keep the vehicle in a straight line and allow the operator to turn if it becomes necessary. The operator can use controlled or stab braking methods. **Note:** If driving a vehicle with anti-lock brakes, read and follow the directions found in the owner's manual for stopping quickly:

5.2.2.1. Controlled braking:

5.2.2.1.1. Apply the brakes as hard as possible without locking the wheels.

5.2.2.1.2. Keep steering wheel movements very small while doing this.

5.2.2.1.3. Release the brakes if it is necessary to make a larger steering adjustment or if the wheels lock and reapply the brakes as soon as possible.

5.2.2.2. Stab braking:

5.2.2.2.1. Apply the brakes all the way.

5.2.2.2.2. Release brakes when wheels lock up.

5.2.2.2.3. As soon as the wheels start rolling, apply the brakes fully again.

5.2.3. Stopping distance. The total stopping distance for vehicles with air brake systems is made up of four different factors. Perception Distance plus Reaction Distance, plus Brake Lag Distance, plus Effective Braking Distance, equals the Total Stopping Distance.

5.2.4. Brake fading or failure:

5.2.4.1. Brakes can fade or fail from excessive heat caused by using them too much and not relying on the engine braking effect. Excessive heat causes chemical changes in the brake lining which reduces friction and causes expansion of the brake drums.

5.2.4.2. Brake fade is also affected by adjustment. When brakes are out of adjustment, they will not be able to share the workload with brakes that are properly adjusted, causing the other brakes to overheat and fade from overuse.

5.2.5. Proper Braking Technique. Once the vehicle is in the proper low gear, follow the proper braking technique: **Note:** The use of brakes on along and/or steep downgrade is only a supplement to the braking effect of the engine.

5.2.5.1. Apply the brakes just hard enough to feel a definite slowdown.

5.2.5.2. When the speed has been reduced to approximately 5 mph below the "safe" speed, release the brakes (for appx. 3 seconds).

5.2.5.3. When the speed has increased to the "safe" speed, repeat the above.

5.2.6. Low Air Pressure:

5.2.6.1. If the low air pressure warning comes on, stop and safely park the vehicle as soon as possible.

5.2.6.2. Spring brakes will engage when the air pressure drops into the range of 20 to 45 psi.

5.2.7. Parking Brakes. To use the parking brake, pull the parking brake control knob out. The control will be a yellow, diamond shaped knob labeled “parking brakes” on newer vehicles. On older vehicles, it may be a round blue knob or some other shape (including a lever that swings from side to side or up and down). **Note:** Never leave the vehicle unattended without applying the parking brakes or chock the wheels. The vehicle might roll away and cause injury and damage.

5.2.7.1. Do not use the parking brakes if the brakes are very hot or they can be damaged (e.g. from just coming down a steep grade). Let hot brakes cool before using the parking

brakes.

5.2.7.2. Do not use the parking brakes if the brakes are very wet in freezing temperatures (vehicle will not be able to move if they freeze). Use wheel chocks to hold the vehicle. If brakes are wet, use the brakes lightly while driving in a low gear to heat and dry them.

5.2.7.3. If the vehicle does not have automatic air tank drains, drain air tanks to remove the moisture and oil.

5.2.8. Safety Reminders.

5.2.8.1. ABS will not prevent power or turning skids.

5.2.8.2. ABS will not increase or decrease ultimate stopping power.

5.2.8.3. ABS will not necessarily shorten stopping distance.

5.2.8.4. ABS will not change the way the vehicle normally brakes.

Chapter 6

EXPLANATION AND DEMONSTRATION

6.1. Trainer Preparation and Safety Protocols. Prior to instruction, the trainer must establish a safe training location, reserve an appropriate wrecker and a training vehicle to be towed, obtain the manufacturer's operator's manual, and ensure the trainee has completed a DAF Form 171. Both trainer and trainee must adhere strictly to safety protocols, including the removal of jewelry and identification tags, the use of wheel chocks when parked, and the mandatory use of seatbelts. The trainer should conduct an initial walk-around to familiarize themselves and the trainee with all warning labels, data plates (specifically noting boom ratings, winch ratings, wire rope specifications, and under-lift capacities), and wrecker dimensions.

6.1.1. Personal Protective Equipment (PPE) and RM. Trainers and trainees must wear appropriate PPE, including safety steel-toed boots, heavy-duty gloves for cargo/chain handling, coveralls, eye/hearing protection, and a reflective vest. Throughout the demonstration, the trainer will actively model the five-step Air Force Risk Management (RM) process, specifically assessing hazards related to suspended loads, highly tensioned cables, pinch points, and active roadway traffic during recoveries.

6.2. Operator Maintenance and Vehicle Pre-Trip Inspection. The trainer will accomplish a comprehensive vehicle inspection alongside the trainee, documenting it on the DAF Form 1800. The trainer will demonstrate the mandated seven-step inspection method (Attachment 4) and utilize the inspection guide (Attachment 2). Crucially, the trainer will demonstrate how to inspect specialized recovery components (winch cables, hooks, pulleys, booms) and, if the wrecker is equipped with air brakes, perform the comprehensive final in-cab air brake tests.

6.3. Operational Demonstration. Utilizing the vehicle data plate and the manufacturer's operator's manual, the trainer will explain the wrecker's capacities, controls, and all warning lights/beacons. The trainer will demonstrate basic unladen driving, including steering, turning (emphasizing the difference in turning radius when towing), changing lanes, and navigating varying grades.

6.3.1. Hook-Up and Recovery Clearances: The trainer must emphasize scene safety, including the placement of warning triangles/flares. The trainer will demonstrate how to properly place the transmission in neutral, engage the Power Take-Off (PTO), and adjust the engine RPM using the throttle control. The trainer will demonstrate proper hook-ups from the curbside to the maximum extent possible, ensuring safety chains are always utilized and the towed load is kept within one foot of the ground.

6.3.1.1. WARNING: The trainer will emphasize critical safety rules: never travel with the PTO engaged, never get under a raised vehicle without jack stands, and never walk beneath a suspended load.

6.3.2. Specialized Systems: Depending on the assigned wrecker type, the trainer will demonstrate the specific recovery process:

6.3.2.1. Flatbed Wreckers: The trainer will demonstrate tilting and extending the bed,

winching the vehicle onto the deck (never driving it on), securing it with safety chains, and retracting the bed.

6.3.2.2. Hook Wreckers: The trainer will demonstrate aligning the tow bar/sling, attaching J-hooks/chains to the axle, securing safety chains, locking spacer bars, and lifting the vehicle. If applicable, the trainer will demonstrate adjusting the split boom, deploying outriggers, and utilizing snatch blocks (two-part/three-part line pulls) and rear scotch blocks.

6.3.2.3. Wheel-Lift Wreckers: The trainer will demonstrate extending the wheel-lift, inserting tire restraints into the cross tube, and tightening safety straps over the tires.

6.3.2.4. Tow Dollies: The trainer will explain when and how to install a tow dolly (e.g., AWD (All-Wheel Drive) vehicles, damaged trailing axles, or when a drive shaft cannot be removed).

6.3.3. Backing Procedures. Due to the size and blind spots of a wrecker (especially with a towed vehicle), Air Force regulations require the use of a spotter. The trainer will review standard hand signals (QTP 24-3) and have the trainee serve as the spotter, demonstrating how to keep the spotter in sight in the mirrors at all times, and stopping the vehicle immediately if visual contact is lost.

6.4. Post-Operation Demonstration. Following the driving and recovery portion, the trainer will demonstrate after-operation procedures. This includes unloading the towed vehicle, properly rewinding the winch cable, stowing chains and straps, disengaging the PTO, cleaning the vehicle, draining the air tanks (mandatory for air brake equipped vehicles), refueling, and executing the manufacturer's shut-down procedures. The trainer will park the vehicle in a level area, apply the parking brake, and perform a post-operation walk-around, annotating any discrepancies on the DAF Form 1800.

Chapter 7

TRAINEE PERFORMANCE DEMONSTRATION AND EVALUATION

7.1. Trainee Performance and Practical Application. Following the demonstration, the trainee will transition to behind-the-wheel practice. The trainer must ensure safety at all times and immediately stop training if any safety infractions occur. If available, the vehicle should be marked with a "Driver in Training" sign. The trainee must don appropriate PPE (steel-toed boots, gloves, reflective vest, eye/hearing protection), fasten their seatbelt, and properly adjust the seat and all mirrors before moving.

7.2. Trainee Vehicle Inspection Execution. The trainee will conduct operator maintenance and pre/during inspections, utilizing **Attachment 2** as a guide. The trainee must verbally explain the items being inspected to the trainer, including the inspection of the winch cable, hooks, and hydraulics. If the vehicle is equipped with air brakes, the trainee must flawlessly execute and explain the in-cab air brake system tests without trainer assistance, ensuring the DAF Form 1800 is properly documented.

7.3. Trainee Operational Execution. The trainer will establish a road course containing turns, intersections, and varying terrain. The trainee will practice operating the wrecker both with and without a towed vehicle until they show proficiency. The trainee must successfully explain the wrecker's load capacities, gauge parameters, and recovery safety techniques.

7.3.1. Operations: They will practice wide turns (accounting for the towed vehicle's swing radius), lane changes, and speed control on upgrades/downgrades. The trainee must demonstrate proper braking compensation for the unladen wrecker versus the heavily loaded wrecker.

7.3.2. Recovery & Systems: The trainee must safely approach a disabled vehicle, secure the scene with warning triangles, and properly engage the PTO and throttle control. Depending on the wrecker equipped, the trainee must flawlessly execute the loading, securing (including two safety chains), and unloading of the training vehicle using the flatbed winch, hook/sling, or wheel-lift system. The trainee must also successfully demonstrate the installation of a wheel dolly and, if applicable, proper snatch block routing. The trainee must disengage the PTO prior to any transit.

7.3.3. Backing Practice. The trainee will practice backing maneuvers utilizing the trainer (or another available trainee) as a spotter. The trainee must strictly adhere to using hand signals, maintaining visual contact via the mirrors, and stopping immediately if the spotter disappears.

7.4. Performance Evaluation. Once proficient, the trainee will complete the official performance evaluation using the checklist found in Attachment 4. The trainer will review the checklist with the trainee and answer any final questions. During the evaluation, the trainer will ensure all safety, PPE, load rating protocols, PTO management, and operator manual warnings are strictly observed. The trainee must complete the evaluation checklist with zero instructor assistance.

7.4.1. Retraining and Re-evaluation. If the trainee fails any portion of the evaluation, they will be issued a "No-Go." The trainer will critique the observed weaknesses and re-demonstrate the failed items. The trainee must re-perform the specific tasks until they show proficiency before an official re-evaluation is conducted.7.4.1. Retraining and Re-evaluation. If the trainee fails any portion of the evaluation, they will be issued a "No-Go." The trainer will critique the observed weaknesses and re-demonstrate the failed items. The trainee must re-perform the specific tasks until they show proficiency before an official re-evaluation is conducted.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 13-213, *Airfield Driving*, 19 November 2023
QTP 24-3, *General Vehicle Safety*, 23 January 2026

Adopted Forms

DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License*, 6 May 2026
DAF Form 623, *Individual Training Record Folder*, 31 March 2023
DAF Form 847, *Recommendation for Change of Publication*, 22 September 2009
DAF Form 1800, *Operator's Inspection Guide and Trouble Report*, 18 February 2026

Abbreviations and Acronyms

ABS—Antilock Brake System
AFIMSC—Air Force Installation Mission Support Center
ASTM—American Society for Testing and Materials
CAC—Common Access Card
CMV—Commercial Motor Vehicle
DAF—Department of the Air Force
GAWR—Gross Axle Weight Rating
GCWR—Gross Combination Weight Rating
GMV—Government Motor Vehicle
GVWR—Gross Vehicle Weight Rating
GVW—Gross Vehicle Weight
IAW—In Accordance With
ITR—Individual Training Record
MPH—Miles per Hour
OJT—On-the-Job Training
PPE—Personal Protective Equipment
PSI—Pounds per Square Inch
PTO—Power Take-off
QTP—Qualification Training Package
RM—Risk Management
RPM—Revolutions per Minute
TO—Technical Order
UTM—Unit Training Manager
VCO—Vehicle Control Official
WLL—Working Load Limit

Attachment 2

WRECKER INSPECTION GUIDE

Figure A2.1. Wrecker Inspection Guide.

Inspection Guide - Wrecker

STEP 1. Vehicle Overview

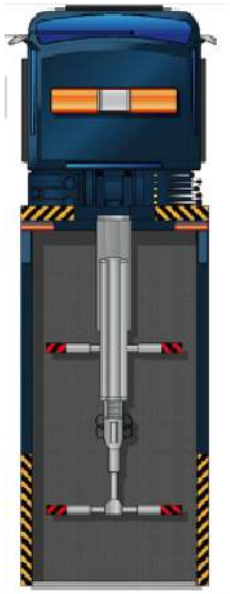
- ☐ **Paperwork**
  - DAF Form 1800
- ☐ **Vehicle Approach**
  - Damage
  - Vehicle Leaning
  - Fluid Leaks (puddling on ground/frame)
  - Surrounding Hazards

STEP 2. Engine Compartment

- ☐ **Leaks/hoses/Electrical Wiring Insulation**
- ☐ **Oil Level**
- ☐ **Coolant Level**
- ☐ **Power Steering Fluid**
- ☐ **Windshield Washer Fluid**
- ☐ **Battery Connections & Tie-downs**
- ☐ **Automatic Transmission Fluid Level**
- ☐ **Engine Compartment Belts**

STEP 3. Engine Start/ Cab Check (Left/Front/Right)

- ☐ **Safe Start**
- ☐ **Gauges**
  - Oil Pressure Gauge
  - Air Pressure Gauge
  - Temperature Gauge (Coolant/Engine Oil)
  - Ammeter/Voltmeter
- ☐ **Warning Lights & Buzzers**
- ☐ **Mirrors & Windshield**
- ☐ **Wipers/Washers**
- ☐ **Emergency & Safety Equipment**
  - Red Reflective Triangles
  - Properly Charged & Rated Fire Extinguisher
  - Optional (Chains/Tire Changing Equip, Emergency Phone List)
- ☐ **Lights/Reflectors/Reflector Tape**
  - Dash Indicators for:
    - Left Turn Signal
    - Right Turn Signal
    - Four-Way Emergency Flashers
    - High Beam Headlight
    - ABS Indicator (If equipped)
    - Clearance Lights
  - Reflective Clean/Functional Light/Reflector Checks Include:
    - Headlights (clear)
    - Taillights (red)
    - Backing Lights (clear)
    - Turn Signals (amber, except in rear)
    - Four-Way Flashers (amber, except in rear)
    - Brake Lights (red)
    - Red Reflectors & Amber Reflectors (side/rear)
    - Reflective Tape Condition
- ☐ **Horn**
- ☐ **Heater/Defroster**
- ☐ **Brakes**
  - Parking Brake Check
  - Hydraulic Brake Check
  - Air Brake Check (if equipped)
  - Service Brake Check
- ☐ **Safety Belt**



STEP 4. Walk-Around Inspection

- ☐ **Steering**
  - Steering Box/Hoses/linkages
- ☐ **Suspension**
  - Springs/Air/Torque/Mounts
  - Shock Absorbers
- ☐ **Brakes**
  - Slack Adjustor's & Push-rods
  - Brake Chambers
  - Brake Hoses/Lines
  - Drum Brake/Brake Linings
- ☐ **Wheels**
  - Rims
  - Tires
  - Hub Oil Seals/Axle Seals
  - Lug Nuts
  - Doors
  - Mirrors
  - Fuel Tank
- ☐ **Recovery Accessories**
  - Air/Electrical lines & connectors
  - Catwalk/Steps
  - Storage Bins
  - Tool Kit
  - Tow light kit (magnet/clip)
  - Straps/Chains/Snatch Blocks/Skids
  - Wheel Dolley
  - Remote control (if equipped)
- ☐ **Recovery Components**
  - Boom
  - Winch
  - Winch Cable (lubricated, frays, etc.)
  - Hooks
  - Wheel Lift
  - Hydraulic cylinders

Attachment 3

PERFORMANCE TEST

A3.1. Desired Learning Outcome.

A3.1.1. Understand the safety precautions to be followed before-, during-, and after-operation of the wrecker.

A3.1.2. Understand the purpose of the wrecker and their role in the mission.

A3.1.3. Know the proper operator maintenance procedures of the wrecker, IAW applicable technical orders and use of DAF Form 1800.

A3.1.4. Safely and proficiently operate the wrecker.

A3.2. Instructions. Before beginning the performance test, the trainer will brief the trainee on the scenario that will need to be accomplished. He/she will be given additional directions and instructions as needed throughout the scenario.

A3.3. Scoring.

A3.3.1. The trainer examiner will be scoring the trainee on wrecker operations and also the general safe driving practices. The examiner will give directions and instructions to the trainee in sufficient time for him/her to execute a driving maneuver. They will not be asked to drive in an unsafe manner.

A3.3.2. The examiner will be making various marks on the performance test checklist. This does not necessarily mean anything has been done wrong. It is in the best interest to concentrate on the operation of the wrecker. The trainer will explain the test results at the conclusion of the performance test.

A3.3.3. Tasks being graded are listed on the following page; the trainee will be required to successfully pass all items.

A3.3.4. The instructor will stop the test at any time safe operations are not being followed or as deemed necessary for safety concerns.

Figure A3.1. Performance Test Checklist.

PERFORMANCE TEST CHECKLIST

Instructions to Certifier and Trainee:

Pre-Test Briefing: The Certifier will review this entire checklist with the trainee and brief the intended road course route. The trainee must understand that all driving directions will be provided in sufficient time to execute maneuvers safely.

Scoring Awareness: The Certifier will make various marks on this checklist during the evaluation. These marks do not inherently indicate a "No-Go"; the trainee must focus exclusively on safe vehicle operation.

Conduct of Test: No Trainer or Certifier assists are permitted during the evaluation. Results will be explained to the trainee upon the conclusion of the test.

Safety: The Certifier will stop the test immediately if unsafe operations occur or health concerns arise.

SECTION I. TRAINEE AND CERTIFIER INFORMATION

1.Trainee Name (Last, First MI)	**2. Unit/Office Symbol**
3. Certifier Name (Last, First MI)	**4. Unit/Office Symbol**

SECTION II. PERFORMANCE TEST INFORMATION

5. Vehicle Management Code for Vehicle Tested	**6. Date**

SECTION III. SCORESHEET

7. Equipment & Posture	Go	No Go
Verifies vehicle required forms & personal protective equipment	☐	☐

8. Pre-Operational & General Safety:	Go	No Go
Conducts 360 degree safety inspection when approaching veh	☐	☐
Inspects items identified on DAF1800, documents discrepancies	☐	☐
If equipped, Air Brake Check	☐	☐
Conducts ops check of controls & instruments	☐	☐
Sign DAF1800	☐	☐
Performs post trip inspection,reports any discrepancies to Veh Mgt	☐	☐

10. Maneuvers (Turns and Curves):	Go	No Go
Turns: Checks traffic in all directions/signals properly.	☐	☐
Turns: Decelerates smoothly/proper gear for the turn.	☐	☐
Turns: Checks mirrors/avoids drifting into oncoming traffic.	☐	☐
Curves: Reduces speed/proper gear prior to entry	☐	☐
Curves: Maintains lane positioning/360-degree traffic checks	☐	☐

9. On-Road Operational Compliance:	Go	No Go
Uses seat belts, obeys traffic signs, signals, & traffic laws.	☐	☐
Completes course without accidents/moving violations.	☐	☐
Regular traffic/mirror checks/safe following distances.	☐	☐
Uses proper signals/smooth, safe lane changes.	☐	☐
Properly utilizes clutch for up-shifting/downshifting.	☐	☐
Doesn't over rev engine, grind gears, or ride the clutch.	☐	☐
Properly utilizes jake brake (when equipped).	☐	☐

11. Intersections and Hazards:	Go	No Go
Int.: Decelerates gently/brakes smoothly on approach.	☐	☐
Int: Stop (no coasting) behind stop lines/signs/sidewalks.	☐	☐
R/R: Decelerates and shifts gears before the crossing.	☐	☐
R/R: Does not stop, shift, or change lanes while crossing.	☐	☐
Signs: Identifies height clearance signs on route	☐	☐

12. Vehicle Recovery Operations:	Go	No Go
Maintains safe operation during recovery (scene awareness, control of equipment, uses chocks, beacons, etc.).	☐	☐
Properly engages/disengages PTO & utilizes throttle control.	☐	☐
Properly uses winching in and out. Ensures tensioned unravel, monitors raveling of winch cable (no bird cage, flat spots, etc.)	☐	☐
Proper use of cable assist products (snatch blocks, out riggers, nylon straps, etc.)	☐	☐
Properly uses wheel lift, securing tire to wheel lift, ensuring gap between power unit and towed asset, and ride height of lift.	☐	☐
Ensures towed asset is properly secured to the wrecker, external lights are attached to tow unit (if necessary), etc.	☐	☐

13. Remarks:

SECTION IV. CERTIFIERS ENDORSEMENT

I, the undersigned, do attest and certify that the trainee named herein has successfully completed all required training and is deemed proficient in the operation of the vehicle(s)/equipment listed in **Section II**. This certification is made to the best of my knowledge and belief, based on training conducted pursuant to the approved qualification training package.

14. Signature	**15. Date**

Attachment 4

SEVEN-STEP INSPECTION GUIDE

Figure A4.1. Seven-Step Inspection Guide.

SEVEN-STEP VEHICLE INSPECTION GUIDE

Step 1: Vehicle Overview

Review Last Vehicle Inspection Report. Drivers may have to make a vehicle inspection report in writing each day. The motor carrier must repair any items in the report that affect safety and certify on the report that repairs were made or were unnecessary. Only sign the previous report after verifying that any noted defects have been repaired or certified as not needing repair..

Step 2: Check Engine Compartment

Parking Brakes Are On and/or Wheels Chocked.

You may have to raise the hood, tilt the cab (ssecure any loose items to prevent them from falling and causing damage), or open the engine compartment door.

Check the following:

- Engine oil level.
- Coolant level in radiator; condition of hoses.

Check That the

- Power steering fluid level; hose condition (if so equipped).
- Windshield washer fluid level

Battery fluid level, connections, and tie downs (battery may be located elsewhere)

- Automatic transmission fluid level (may require engine to be running).
- Check belts for tightness and excessive wear (alternator, water pump, air compressor) --learn how much "give" the belts should have when adjusted right and check each one.
- Leaks in the engine compartment (fuel, coolant, oil, power steering fluid, hydraulic fluid, battery fluid).
- Cracked, worn electrical wiring insulation.
- Lower and secure hood, cab, or engine compartment door.

Step 3: Start Engine and Inspect Inside the Cab

Get In and Start Engine

Make sure parking brake is on.

Put gearshift neutral (or "park" if automatic).

Start engine; listen for unusual noises.

If equipped, check the Anti-lock Braking System (ABS) indicator lights. Light on dash should come on and then turn off. If it stays on the ABS is not working properly. For trailers only, if the yellow light on the left rear of the trailer stays on, the ABS is not working properly.

Look at the Gauges

Oil pressure. Should come up to normal within seconds after engine is started. See Figure 2.5

Air pressure. Pressure should build from 50 to 90 psi within 3 minutes. Build air pressure to governor cut-out (usually around 120 – 140 psi. **Know your vehicle's requirements.**

Ammeter and/or voltmeter. Should be in normal range(s).

Coolant temperature. Should begin gradual rise to normal operating range.

Engine oil temperature. Should begin gradual rise to normal operating range.

Warning lights and buzzers. Oil, coolant, charging circuit warning, and antilock brake system lights should go out right away.

Check Condition of Controls. Check all of the following for looseness, sticking, damage, or improper setting:

- Steering wheel.
- Clutch.
- Accelerator ("gas pedal").
- Brake controls.
- Foot brake.
- Trailer brake (if vehicle has one).
- Parking brake.
- Retarder controls (if vehicle has them).
- Transmission controls.
- Interaxle differential lock (if vehicle has one).
- Horn(s).
- Windshield wiper/washer.
- Lights.
- Headlights.
- Dimmer switch.
- Turn signal.
- Four-way flashers.
- Parking, clearance, identification, marker switch(es).

OIL PRESSURE

- Idling 5-20 PSI
- Operating 35-75 PSI
- Low, Dropping, Fluctuating: **STOP IMMEDIATELY!** Without oil the engine can be destroyed rapidly

Figure 2.5

Check Mirrors and Windshield. Inspect mirrors and windshield for cracks, dirt, illegal stickers, or other obstructions to seeing clearly. Clean and adjust as necessary.

Check Emergency Equipment

Check for safety equipment:

Spare electrical fuses (unless vehicles have circuit breakers).

Three red reflective triangles, 6 fuses or 3 liquid burning flares.

Properly charged and rated fire extinguisher.

Check for optional items such as:

- Chains (where winter conditions require).
- Tire changing equipment.
- List of emergency phone numbers
- Accident reporting kit (packet).

Check Safety Belt. Check that the safety belt is securely mounted, adjusts; latches properly and is not ripped or frayed.

Step 4: Turn Off Engine and Check Lights

Make sure the parking brake is set, turn off the engine, and take the key with you. Turn on headlights (low beams) and four-way emergency flashers and get out of the vehicle.

Step 5: Do Walk-around Inspection

- Go to front of vehicle and check that low beams are on, and both of the four-way flashers are working.
- Push dimmer switch and check that high beams work.
- Turn off headlights and four-way emergency flashers.
- Turn on parking, clearance, side-marker, and identification lights.
- Turn on right turn signal and start walk-around inspection.

General

- Walk around and inspect.
- Clean all lights, reflectors, and glass as you go along.

Left Front Side

- Driver's door glass should be clean.
- Door latches or locks should work properly.
- Left front wheel.
- Condition of wheel and rim--missing, bent, broken studs, clamps, lugs, or any signs of misalignment.
- Condition of tires--properly inflated, valve stem and cap OK, no serious cuts, bulges, or tread wear.
- Use wrench to test rust-streaked lug nuts, indicating looseness.
- Hub oil level OK, no leaks.
- Left front suspension.
- Condition of spring, spring hangers, shackles, U-bolts.
- Shock absorber condition.
- Left front brake.
- Condition of brake drum or disc.
- Condition of hoses.

Front

- Condition of front axle and steering system.
- No loose, worn, bent, damaged or missing parts.
- Must grab steering mechanism to test for looseness.
- Condition of windshield.
- Check for damage and clean if dirty.
- Check windshield wiper arms for proper spring tension.
- Check wiper blades for damage, "stiff" rubber, and securement.
- Lights and reflectors.
- Parking, clearance, and identification lights clean, operating, and proper color (amber at front).
- Reflectors clean and proper color (amber at front).
- Right front turn signal light clean, operating, and proper color (amber or white on signals facing forward).

Right Side

- Right front: check all items as done on left front.
- Primary and secondary safety cab locks engaged (if cab-over-engine design).
- Right fuel tank(s).
- Securely mounted, not damaged, or leaking.
- Fuel crossover line secure.
- Tank(s) contain enough fuel.
- Cap(s) on and secure.
- Condition of visible parts.
- Rear of engine--not leaking.
- Transmission--not leaking.
- Exhaust system--secure, not leaking, not touching wires, fuel, or air-lines.
- Frame and cross members--no bends or cracks.
- Air-lines and electrical wiring--secured against snagging, rubbing, wearing.
- Spare tire carrier or rack not damaged (if so equipped).
- Spare tire and/or wheel securely mounted.
- Spare tire and wheel adequate (proper size, properly inflated).
- Cargo securement (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Header board adequate, secure
- Side boards, stakes strong enough, free of damage, properly set in place
- Canvas or tarp properly secured to prevent tearing, billowing, or blocking of mirrors.
- If oversize, all required signs (flags, lamps, and reflectors) safely and properly mounted and all required permits in driver's possession.
- Curbside cargo compartment doors in good condition, securely closed, latched/locked and required security seals in place.

Right Rear

- Condition of wheels and rims--no missing, bent, or broken spacers, studs, clamps, or lugs.
- Condition of tires--properly inflated, valve stems and caps OK, no serious cuts, bulges, tread wear, tires not rubbing each other, and nothing stuck between them.
- Tires same type, e.g., not mixed radial and bias types.
- Tires evenly matched (same sizes).
- Wheel bearing/seals not leaking.
- Suspension.
- Condition of spring(s), spring hangers, shackles, and U-bolts.
- Axle secure.
- Powered axle(s) not leaking lube (gear oil).
- Condition of torque rod arms, bushings.
- Condition of shock absorber(s).
- If retractable axle equipped, check condition of lift mechanism. If air powered, check for leaks.
- Condition of air ride components.
- Brakes.
- Brake adjustment.
- Condition of brake drum(s) or discs.
- Condition of hoses--look for any wear due to rubbing.
- Lights and reflectors.
- Side-marker lights clean, operating, and proper color (red at rear, others amber).
- Side-marker reflectors clean and proper color (red at rear, others amber).

Rear

- Lights and reflectors.
- Rear clearance and identification lights clean, operating, and proper color (red at rear).
- Reflectors clean and proper color (red at rear).
- Taillights clean, operating, and proper color (red at rear).
- Right rear turn signal operating, and proper color (red, yellow, or amber at rear).
- License plate(s) present, clean, and secured.
- Splash guards present, not damaged, properly fastened, not dragging on ground, or rubbing tires.
- Cargo secure (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Tailboards up and properly secured.
- End gates free of damage, properly secured in stake sockets.
- Canvas or tarp (if required) properly secured to prevent tearing, billowing, or blocking of either the rearview mirrors or rear lights.
- If over-length, or over-width, make sure all signs and/or additional lights/flags are safely and properly mounted and all required permits are in driver's possession.
- Rear doors closed, latched/locked.

Left Side

- Check all items as done on right side, plus:
- Battery(ies)
- Battery box(es) securely mounted.
- Box has secure cover.
- Battery(ies) secured against movement.
- Battery(ies) not broken or leaking.
- Fluid in battery(ies) at proper level (except maintenance-free type).
- Cell caps present and securely tightened (except maintenance-free type).
- Vents in cell caps free of foreign material (except maintenance-free type).

Step 6: Check Signal Lights

- Get In and Turn Off Lights
- Turn off all lights.
- Turn on stop lights (apply trailer hand brake or have a helper put on the brake pedal).
- Turn on left turn signal lights.
- Get Out and Check Lights
- Left front turn signal light clean, operating and proper color (amber or white on signals facing the front).
- Left rear turn signal light and both stop lights clean, operating, and proper color (red, yellow, or amber).

Get In Vehicle

- Turn off lights not needed for driving.
- Check for all required papers, trip manifests, permits, etc.
- Secure all loose articles in cab (they might interfere with operation of the controls or hit you in a crash).
- Start the engine.

Step 7: Start the Engine and Check

Test for Hydraulic Leaks. If the vehicle has hydraulic brakes, pump the brake pedal three times. Then apply firm pressure to the pedal and hold for five seconds. The pedal should not move.

If it does, there may be a leak or other problem. Get it fixed before driving. If the vehicle has air brakes, do the checks described in Sections 5 and 6 of this manual.

Brake System

Test Parking Brake(s)

- Fasten safety belt
- Set parking brake (power unit only).
- Release trailer parking brake (if applicable).
- Place vehicle into a low gear.
- Gently pull forward against parking brake to make sure the parking brake holds.
- Repeat the same steps for the trailer with trailer parking brake set and power unit parking brakes released (if applicable).
- If it doesn't hold vehicle, it is faulty; get it fixed.

Test Service Brake Stopping Action

- Go about five miles per hour.
- Push brake pedal firmly
- "Pulling" to one side or the other can mean brake trouble.
- Any unusual brake pedal "feel" or delayed stopping action can mean trouble.
- If any defects are found during the inspection, they must be repaired before operating the vehicle. Federal and state laws prohibit driving an unsafe vehicle.